

Optimization of the configuration and flexible operation of the pipe-embedded floor heating with low-temperature district heating



Xiaochen Yang^{a,b,c,*}, Li Pan^{a,b,c}, Wenze Guan^{a,b,c}, Zhe Tian^{a,b,c,*}, Jing Wang^d, Chen Zhang^e

^aTianjin Key Lab of Indoor Air Environmental Quality Control, School of Environmental Science and Engineering, Tianjin University, Tianjin, China

^bTianjin Key Laboratory of Built Environment and Energy Application, Tianjin University, Tianjin 300072, China

^cMOE Key Laboratory of Efficient Utilization of Low and Medium Grade Energy, Tianjin University, Tianjin 300072, China

^dDalian Architecture Science Research and Design Institute CO., LTD, Dalian 116014, China

^eDepartment of the Built Environment, Thomas Manns Vej 23, 9220 Aalborg Ø, Denmark

ARTICLE INFO

Article history:

Received 13 April 2022

Revised 20 May 2022

Accepted 9 June 2022

Available online 13 June 2022

Keywords:

Low-temperature district heating

Radiant floor heating

Modelica models

Heat flexibility potential

ABSTRACT

The radiant heating is cost effective way of space heating while providing favourable thermal comfort. Due to the large thermal flexibility, it is possible to utilize the pipe embedded radiant heating for low-temperature heating and demand management, thereby giving more access to the low-grade heat sources such as the renewable energy or the industrial waste heat. In this study, we investigate the optimal configuration of the radiant floor heating system considering different district heating (DH) scenarios. We propose the scheduled control strategy considering the comfort requirements, the occupancy and the dynamic energy price. Parametric simulations regarding the pipe spacing, the thickness of the concrete core, and the DH supply temperatures are performed by Modelica models. The critical levels of the radiant floor configurations of applying medium-temperature DH (MTDH), low-temperature DH (LTDH) and ultra-low temperature DH (ULTDH) are investigated. The indoor comfort, the thermal performances, the economy performances and the energy flexibility as the crucial factors of different cases are analysed and compared. The results show that the pipe spacing of 300 mm achieves better comfort and shorter charging time for all DH scenarios of the case nearly zero energy building (NZEB). The larger thickness of the concrete core can stabilize the indoor temperature variation and buffer the impact from the environmental disturbance more effectively.

© 2022 Elsevier B.V. All rights reserved.

1. Introduction

Along with the development of the urbanization and economy boost, the energy consumed along the building construction and operation process becomes the largest part of the global final energy consumption, which was 36% in 2020 according to the IEA report [1,2]. The accompanied CO₂ emission reaches as much as 37%, which makes the building sector the primary problematic sector for greenhouse gas (GHG) mitigation. To meet the decarbonization goal worldwide, the indoor heating and cooling system with higher energy efficiency and lower CO₂ emission should be paid more attention.

The radiant heating/cooling system is considered as an efficient way of providing indoor comfort due to the operative temperature level. It is considered as the good application for high-temperature

cooling and low-temperature heating. The supply temperature range is normally 25–40 °C for heating and 16–20 °C for cooling [3–5]. Therefore, the radiant heating/cooling system is more suitable for the low-grade renewable heat sources and the waste heat sources [6]. Since the building energy demand is low-exergy demand, the radiant energy supply system reaches higher exergy efficiency in general [7,8]. Moreover, the radiant heating/cooling system are believed to provide better indoor temperature distribution and avoid typical draught problems by the convective system [9,10].

The efficiency of the radiant heating/cooling system was compared to the conventional system considering different parameters. Dréau and Heiselberg compared the performances of four types of indoor heating and cooling terminals by parametric simulations, and found the convective terminals are less efficient compared to the radiant terminals due to the dependency on air change rate and outdoor temperature [11]. Mikeska and Svendsen found the indoor comfort temperature can be provided by the radiant cooling wall with only 4 K temperature difference, which results

* Corresponding authors.

E-mail addresses: xiaochen.yang@tju.edu.cn (X. Yang), tianzhe@tju.edu.cn (Z. Tian).

in higher efficiency of the cooling system [12]. Liao and et al. investigated the energy performances of the radiant heating ceiling in a few sites with different functionalities, and find the energy consumption of the building implemented the radiant ceiling heating is 30% less compared to the standard heat index [13]. Kazanci and Shukuya compared the exergy performances of different heating terminals, and found the renewables combined radiant system has higher exergy efficiency up to 10.5% [7]. Larwa and et al. found the location of phase change material (PCM) layer can significantly influence the thermal performance of the radiant floor heating with wet sand as screed material [14].

In fact, the structure and configuration of the radiant heating/cooling system can significantly affect the performances of the heat transfer process. Ning and et al. found the concrete thickness, pipe spacing and concrete type play a role in the response time [15]. Krajčák and et al studied the effect of pipe spacing, insulation thickness, and core thickness by parametric analysis among three different configurations of radiant cooling walls. They defined the heat storage efficiency coefficient and compared its performance in predicting the thermal response with the time constant and response time [16]. Yang and et al. performed uncertainty and global sensitivity analysis on the floor structural parameters, and found the conductivity of the embedded pipe is more critical structure parameter [17]. Wang D and et al. investigated the intermittent floor heating process by numerical simulation and found that the pipe spacing had an impact on the preheating time and the thickness of the filling layer had a significant effect on the heat release time [18]. Ding P and et al. used a simplified calculation method to research the effect of capillary radiant heating floor on heat transfer performance and found that the heating capacity gradually decreases as the pipe spacing increases [19]. Jing H and et al. proposed two new radiant floor heating structures of cement tile flooring and keel wood flooring and found that for every 10 mm increase in the thickness of the filling layer, the floor surface temperature decreased by 0.5 °C [20]. J. Oravec and et al. found that if a metal fin with high conductivity is added between the pipe and the concrete layer to enhance heat transfer, it would have a great improvement on the thermal output, storage capacity and controllability of the radiant floor [21]. Dong and et al. combined the advantages of conventional finned tube heat exchanger based on forced convective heat transfer and the direct radiant heat terminal for a new radiant-convective heating system, which have verified the feasibility of the new heating system [22].

Because of the large thermal capacity, the radiant heating/cooling system can be operated for the aim of the load shifting or peaking shaving of the energy system [23,24]. However, the high thermal inertia in the other hand results in long time delay for the heating/cooling procedure, thereby hampers the interaction between the indoor environment and the energy supply. In order to achieve the better performance of the radiant heating/cooling system in a particular situation, the characteristics of the heat transfer and thermal inertia of the radiant slab should be considered. Radiant floor heating systems have a high thermal inertia owing to heavy capacitance of the floor structure mass, so it is possible to save energy and maintain thermal comfort temperatures by establishing suitable control methods [25]. A two-dimensional numerical analysis of a radiant floor heating system with intermittent operation was performed by Zhang L and et al. The results revealed the intermittent operation has a significant impact on energy-saving potential and thermal inertia [26].

The rule-based control (RBC) is widely used in the building heating and cooling strategy due its simplicity. Li and et al. found the traditional thermostatic valve with PID control strategy shows few effects on mitigating the overheating of the radiant heating system exposed directly to the solar radiation [27]. Guo and et al.

tested the performances of the ventilation assisted floor heating system by different control strategies, and found the solar radiation can increase the indoor air temperature by 3 °C [28]. Le Dréau J and et al. took into account energy and thermal comfort to develop simple control strategy in underfloor heating to realize the potential of energy flexibility [29]. Luc KM and et al. studied the energy flexibility potential of a small district connected to a DH network by using the thermal mass of the building as the heat storage based on schedule control. The results show that the load can be shifted by establishing a suitable control strategy [23]. Foteinaki K and et al. analyzed thermal storage capacity existing inherently in the building mass by adding or curtailing energy over a period of time without violating thermal comfort, and the results of the study showed that low-energy buildings could be kept within the thermal comfort temperature range for several hours [30].

In recent years, the DH system is developing towards lower temperature so to improve the efficiency of the energy source utilization by making use of the low-grade heat that will be otherwise wasted [31–33]. The radiant heating system is promising solution if combined with the LTDH system due to the better exergy efficiency and large heat flexibility. However, the investigation of the optimal configuration of the hydronic radiant floor heating within the LTDH context has not been sufficient documented based on the author's knowledge. Therefore, this study aims at identifying the crucial configuration parameters of applying the radiant heating under different DH supply temperature levels. The pipe spacing and the thickness of the filling layer as the two important factors were considered. A measured Danish NZEB assisted with heat supply was applied as the reference case. A schedule-based control strategy that integrates the occupancy, the dynamic energy price and the indoor thermal comfort is developed to compare the system performances with the different configurations for load shifting. The different supply temperatures of the MTDH, LTDH, and the ULTDH were applied as boundary conditions. With a reasonable radiant floor heating system configuration, new control strategy can be used to completely shift peak load and save costs within the indoor thermal comfort temperature range.

2. Methods

The configuration of the radiant floor plays an important role in the heat transfer performance of the indoor heating process, which limits the supply temperature of the DH scenario to be applied. This chapter describes the methods of identifying the critical configuration parameters of the different DH scenarios, which includes four main parts, the system model, the control logic, the investigated configuration parameters and the evaluation metrics. Full scale measurements were performed in a Danish NZEB which uses radiant floor heating. Accordingly, a dynamic model is established for the parametric simulations. A well scheduled control strategy is developed to compare the performances of the different scenarios under the load shifting process.

This section is structured as follows. Section 2.1 is the description of the case study and the system model, introducing the components of the system and the settings of parameters according to local code requirements. Section 2.2 develops a new heating strategy based on the occupancy, the dynamic electricity prices and the indoor comfort temperature range. Section 2.3 describes the parameters of the simulated cases, including the DH primary supply temperature, the pipe spacing and the filling layer thickness. Section 2.4 presents the major evaluation indicators of the flexibility provided by the proposed system in terms of time, energy, and economy.

2.1. Case study and on-site measurements

The single-storey house is located in Aarhus, Denmark, where the room has the total area of 132 m² and the treated volume of 360.6 m³. The floor plan of the house is shown in Fig. 1. But the users mainly use the living room, master bedroom, dressing room and bathroom 1, of which the total area is 44.9 m². So the total heat load is calculated based on the heating area in use. According to the Danish energy labelling standard [34], the house is a near-zero energy building house. The parameters of the building envelope are shown in Table 1.

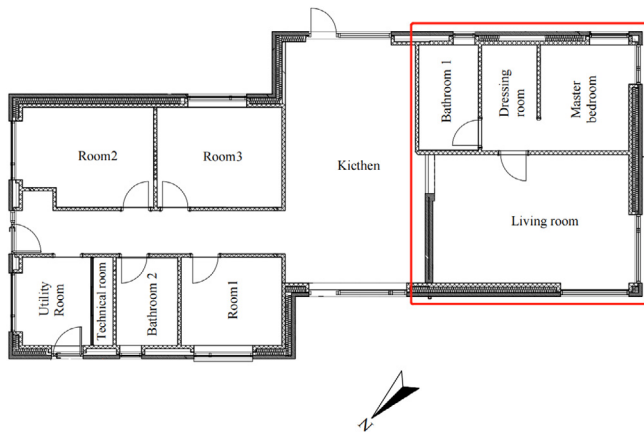


Fig. 1. Floor plan of the case building.

Table 1
Physical parameters of the building envelope.

Building components	U (W/m ² K)	Area (m ²)	Materials
External walls	0.15	209	An outer layer of facade stone and an internal layer of aerated concrete blocks with a thermal insulation layer interposed.
Ground slab	0.08	132	Concrete, with 32 cm of EPS and wooden floor.
Flat roof	0.09	132	Reticular beams, OSB panels and 41 cm of paper wool insulation.
External windows	0.9	23.4	48 mm triple pane glazing with argon, and composite aluminum and wood frames with thermal break.

The radiant floor structure schematic diagram is shown in Fig. 2. The surface layer is made of wood. The filling layer where the embedded pipe is located is made of concrete. The insulation layer is made of expanded polystyrene board (EPS) to prevent heat transfer between the floor and the ground. The DN25 heat pipe in the form of a serpentine is installed in the case study, which is widely used for the pipe embedded radiant floor. The detailed information of each floor layer from top to bottom is shown in Table 2, the information is used as input parameters in the model.

The simulation model consists of four key parts: the room model, the indoor temperature control, the substation model and the heat source model, as shown in Fig. 3. The Room model is simulated by the MixedAir model of the Buildings library in Modelica. The weather data are referenced from the typical meteorological year in Denmark provided by Energy Plus weather website [35]. The proposed model simulated the two weeks containing the coldest day from Dec. 4th to Dec. 17th as shown in Fig. 4. The ventilation rate of the building is assumed to be 0.3 l/s per m² according to the standard [36]. The annual average soil temperature in Denmark is 8 °C calculated as the average value at the depth of 0.5 m from the ground [37]. According to DS/EN ISO 13790:2008, Appendix G.8 [38], it is assumed that convection and radiation have the same proportion of the total internal heat gain from the occupants and equipment, which is 5.8 W/m².

The carried heat from the heat source passes towards upper and bottom directions respectively. To the upper direction, the heat carrier inside the tube transfers heat to the floor surface through the pipe wall, the filling layer and the wooden surface layer. In the radiant slabs model, the connector of the floor upper surface is directly connected to the surf_conBou of the Room model. Towards the bottom direction, the heat from the hot water tends to pass to the ground through the insulation layer, but the insulation layer reduces the heat transfer significantly due to the small thermal conductivity. The radiant slabs on the bottom surface of the insulation layer is still connected to the soil, since small heat loss cannot be neglected due to the temperature difference between the soil and the bottom layer of the floor.

The substation is indirect in-building substation, which connects the primary side and the secondary side through the plate heat exchanger. The heat exchange efficiency is assumed to be 95% assuming it is well insulated. The air source heat pump is assumed to be the main heat source of the DH system, which has the flexibility to reduce the DH supply temperature level to LT_{DDH} and UL_{TDH}. In Denmark, several demonstrated heat pumps supplied DH systems are operated under variable temperatures

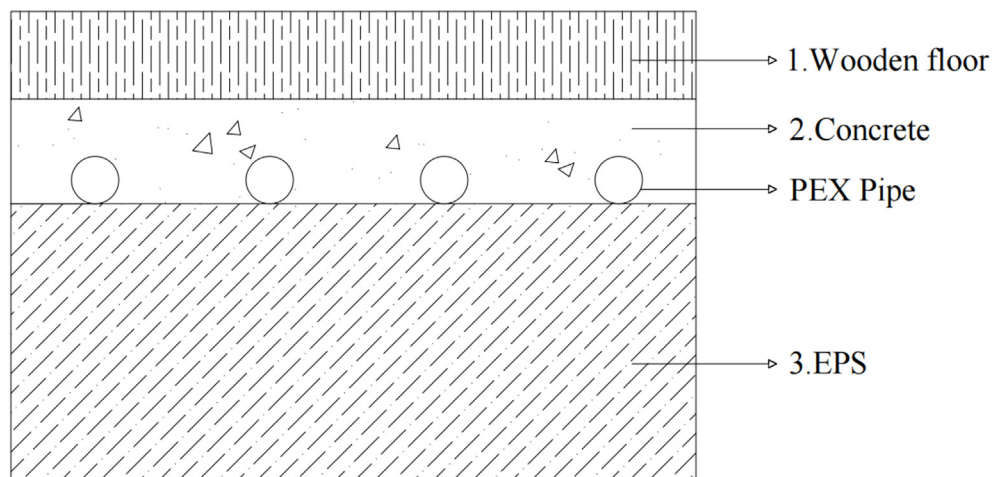


Fig. 2. Materials and layouts of the radiant floor.

Table 2
Materials and properties of the radiant floor from top to bottom layer.

Materials	Thermal conductivity (W/(m·K))	Thickness (mm)	Density (kg/m ³)	Heat capacity (J/(kg·K))
Floor cover -wood	0.18	20	686	1775
Middle layer - Concrete	1.51	120	2300	920
Water pipe -PEX	0.45	1.5	983	–
Bottom layer -EPS	0.033	320	25	1286

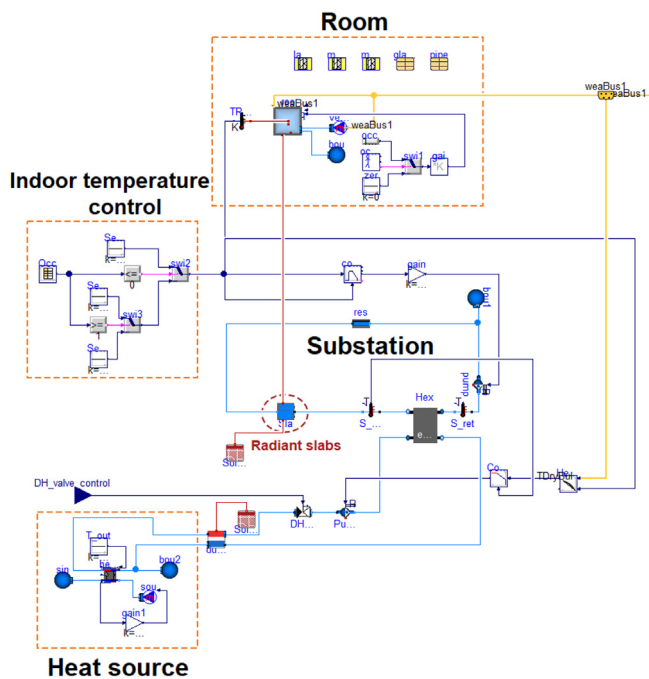


Fig. 3. Schematic diagram of the Modelica model.

according to the specific control strategy. The data are referenced from the product catalog provided by the manufacturer [39]. At 7 °C ambient temperature and 45 °C water outlet temperature under nominal working condition, the heating power of the heat pump is 22 kW, the input power is 6567 W with the COP value of 3.35. The heat pump is switched on and off by a variable frequency valve (DH valve). The DH mass flowrate is regulated by the PID controller on the secondary side, so to minimize the differ-

ence between the required and the actual supply heat. The outdoor design temperature for the weather compensation is set at −12 °C [40]. However, the solar radiation, equipment and internal heat gain will compensate for heat transfer losses, so the weather compensation curve was offset by 8 °C in the model, which means that the heating system is turned off when the outdoor air temperature is higher than the set temperature minus 8 K. The scheduled control logic of the indoor operative temperature settings was carried out using the StateGraphRoot component in Dymola.

2.2. Set temperature and control strategy

The scheduled control strategy of the setpoint room temperature considers the room occupancy, the indoor comfort temperature category and the dynamic electricity price of the heat pump. Fig. 5 shows the average hourly CO₂ concentration variation in the bedroom of the two-week test period. The CO₂ concentration starts to decrease gradually when the users leave home at about 8:00 am on weekdays and increases gradually when they return at 18:00 pm. So the room occupancy was assumed to last from 18:00 pm until 8:00 am the next day. The dynamic electricity price was derived from the Nordpool historical data [41]. The average dynamic hourly electricity price is calculated as the average of the highest and the lowest electricity prices in an hour. In Fig. 6, we can find that the high electricity price periods (above the average value) are from 7:00 to 13:00 and 17:00 to 23:00, and the rest of the time period is low-price period (below the average value).

Considering the small difference between the indoor air temperature and the operative temperature in the case study, as well as the complexity of data sensing and control in practice, we use the indoor air temperature instead of the operative temperature for simplification. The thermal comfort temperature range in Denmark is divided into three categories according to the Danish standard DS-EN15251 [42], as shown in Table 3. In order to make full use of the available heating power during low-price periods

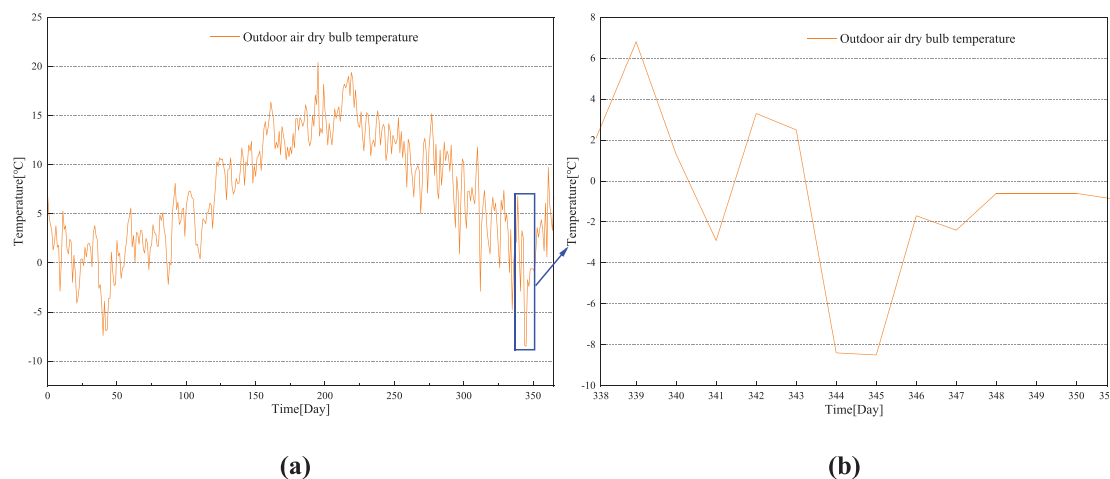


Fig. 4. Outdoor dry bulb temperature variation of the typical meteorological year in Denmark (a) and the zoom of the temperature variation during the coldest two weeks from December 4th to December 17th (b).

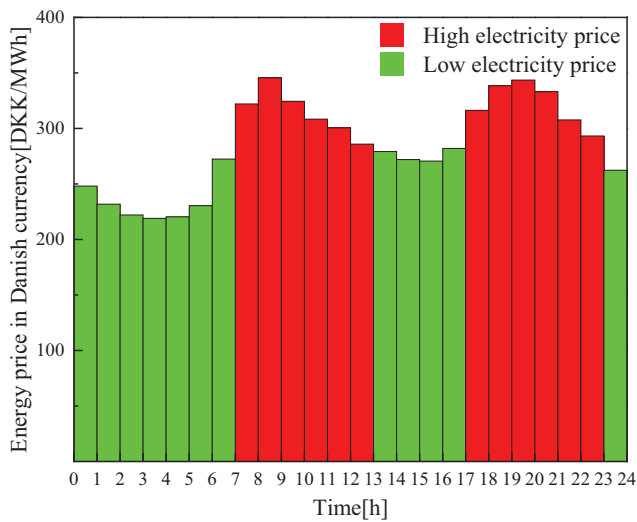


Fig. 5. Hourly average CO₂ concentration in the master bedroom from December 4th to December 17th.

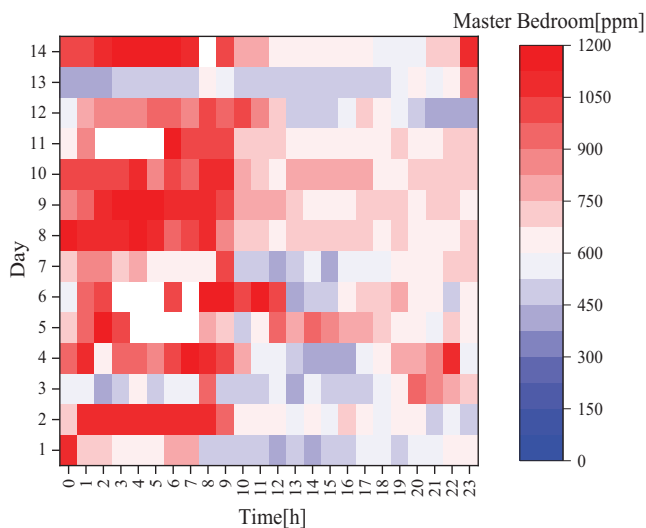


Fig. 6. Average hourly electricity price in Denmark in 2019.

Table 3

Three comfort categories in residential buildings.

Category	Indoor air temperature range (°C)
I	21 ~ 25 °C
II	20 ~ 25 °C
III	18 ~ 25 °C

and to meet the optimal thermal comfort during the occupancy period, we set the room dynamic temperature as shown in Fig. 7.

We propose a scheduled heating strategy, which considers the room occupancy, the variable electricity price and the thermal comfort temperature. As shown in Fig. 7, during the period from 0:00 to 7:00 when the electricity price belongs to the low-price period and the house is occupied, the DH valve is turned on to when the room temperature is below 22 °C using the low-price heat. During the period from 7:00 to 8:00 when the electricity price belongs to the high-price period and the house is occupied,

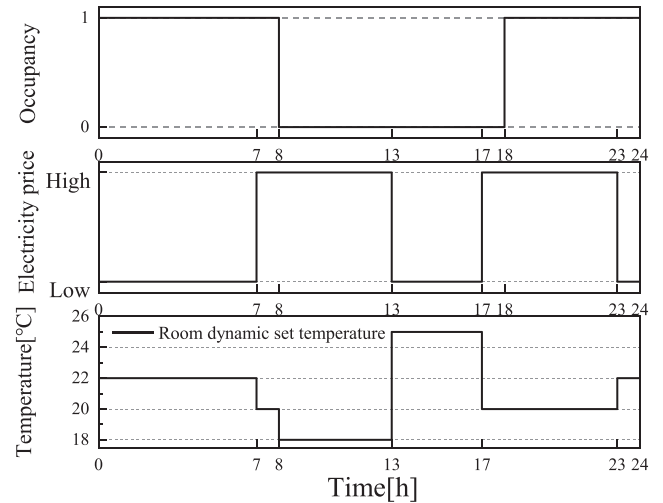


Fig. 7. Room occupancy, electricity price category and the scheduled indoor setpoint temperature.

If the room temperature is below 20 °C, the DH valve will be turned on and maintained at 20 °C for emergency security. From 8:00 to 13:00 when the electricity price is high and the room is unoccupied, the DH valve is switched off unless the room temperature drops below 18 °C. From 13:00 to 17:00, the DH valve is switched on again to charges the building mass of the house to 25 °C and maintains it until 17:00, so that to make use of the cheap heat supply as much as possible. From 17:00 to 18:00, the DH valve is switched off to avoid expenses caused by high-price supplied heat, and the setpoint room temperature is reduced to 22 °C. During the period from 18:00 to 23:00, when the electricity price belongs to the high period and the house is occupied, the room temperature is maintained by the heat released from the building mass. If the room temperature is lower than 20 °C, the DH valve is opened for emergency. From 23:00 to 0:00, the house is heated by DH and the room temperature is maintained at 22 °C instead of 25 °C. The reason is that the room will be overheated due to the large thermal inertia of the building mass, even though the DH valve closes when the room temperature reaches 25 °C. On the other hand, the room temperature was not set to the upper boundary of the thermal comfort in order to take advantage of the heat stored during the preheating period to reduce unnecessary heat loss.

2.3. Parametric analyses of the radiant floor configuration

The pipe spacing and the filling layer thickness are two of the most significant factors that affect the thermal performance of the radiant floor heating system. Therefore, parametric analyses were performed regarding the two factors with the proposed control strategy. The optimal settings of the floor heating structure under different DH supply temperature levels were identified, so to achieve the purpose of reducing energy cost and cutting peak value on the premise of satisfying users' thermal comfort. DH supply temperatures of 35 °C, 45 °C, 50 °C and 60 °C were assumed to represent the MTDH, LTDH, and ULTDH scenarios.

We firstly investigated the critical pipe spacing value for different DH scenarios. The system performances including the deviation between indoor actual temperature and setpoint temperature were compared for 5 pipe spacing levels (100 mm, 200 mm, 300 mm, 400 mm and 500 mm) with assuming the pipe diameter and the filling layer thickness as constants (DN25 and 120 mm respectively). Afterward, the optimal thickness of the

filling layer is analysed. The minimum thickness of the filling layer is set to 50 mm to meet the minimum strength requirement and to maintain the uniform heat distribution on the floor surface. Based on that, we investigated nine levels of the thickness for comparison, which are 50 mm, 70 mm, 90 mm, 110 mm, 120 mm, 130 mm, 150 mm, 170 mm and 190 mm. The effect of the thickness of the filling layer on the thermal and economic performances of the nine cases were compared.

2.4. Evaluating indicator

2.4.1. Thermal comfort

The relative time (RT) of comfort is used to describe the capability of different floor configurations in maintaining the user's thermal comfort with insufficient heat supply.

$$RT = \frac{\tau_1}{\tau_2} \quad (1)$$

where,

τ_1 is the accumulated period when the room temperature meets the comfort category I and the occupancy = 1.

τ_2 is the total time length during the test when the occupancy = 1.

RT is used to indicate the degree to which the room temperature is maintained within the dynamic comfortable range. The value range of RT is [0–1]. The closer it approaches to 0, the weaker the capability of the building mass to maintain room comfort temperature. On the contrary, RT closes to 1 means the building mass has the strong capability in maintaining the indoor comfort temperature.

2.4.2. Heat flexibility of the radiant floor

The energy flexibility of a building is defined as “the ability to manage its demand and generation according to local climate conditions, user needs, and energy network requirements” in IEA EBC Annex 67 [43]. The charging/discharging rates of the building mass are important parameters of characterising the thermal behaviour. In this paper, the charging rate of the floor heating is calculated according to the energy balance equation Eq. (2). $\dot{Q}_{cr}$ describes the rate of the heat flowrate from the heat source charging the radiant floor, which is affected by a few parameters such as the supply temperature, the thermal resistance, the heat pipe arrangement and etc. While $\dot{Q}_{dr}$ describes the rate of the heat flowrate from the radiant floor discharging the room air. In the dynamic simulation, $\dot{Q}_{cr}$ is always greater than 0, and the $\dot{Q}_{dr}$ is smaller than 0.

$$\begin{cases} \dot{Q}_{cr} = \dot{Q}_{sla} - \dot{Q}_{shtr} - \dot{Q}_{loss} > 0 \\ \dot{Q}_{dr} = \dot{Q}_{sla} - \dot{Q}_{shtr} - \dot{Q}_{loss} < 0 \end{cases} \quad (2)$$

where,

$\dot{Q}_{shtr}$ is the surface heat transfer rate, which is the heat flow from the floor to the room (kW).

$\dot{Q}_{loss}$ is the heat loss from the floor to the soil (kW).

$\dot{Q}_{sla}$ is the heat flow from the heat carrier in embedded pipe to the radiant slabs (kW).

$$\dot{Q}_{sla} = m_{sla} * C_{water} * (T_{sup} - T_{ret}) \quad (3)$$

where,

m_{sla} is the mass flow rate through the radiant slabs (kg/s).

C_{water} is the specific heat capacity of water 4.2 kJ/(kg·K).

T_{sup} is the temperature of the water supply for radiant slabs (°C).

T_{ret} is the temperature of the water return for radiant slabs (°C).

Being seen as battery, the actual thermal energy that is stored in the radiant floor determines the time length of the room temperature that can be maintained without exogenous heat input, thereby influencing the control strategy. The heat of the preheat time is mainly used for load shifting at the high price period, so when calculating the thermal storage capacity (which is also regarded as the heat flexibility potential) only considered the unoccupied period. The heat flexibility potential is calculated as the time integral of $\dot{Q}_{cr}$ during the charging period if the room is unoccupied, as shown in Eq.(4).

$$Q_{hfp} = \int_{\tau_3}^{\tau_4} \dot{Q}_{cr} dt \quad (4)$$

where,

τ_3 is the starting time of the charging process when the occupancy = 0.

τ_4 is the ending time of the charging process when the occupancy = 0.

Q_{hfp} is storing heat in the radiant floor, which can be used for load shifting (kWh).

2.4.3. Economic indicators and electricity cost

In order to maximize the usage of the low-price energy, the flexibility factor (FF) is employed to evaluate the capability of the radiant floor for load shifting from high to low prices [28]. The value of FF ranges from –1 to 1. FF = 1 is the optimal case meaning that the entire energy can be shifted to the low-price period. The opposite is FF value below 0, which means the building mass has limited flexibility for shifting the heat load to low-price period. The extreme case is FF = -1, which means all energy consumption occurs during the high price period, but the system has little flexibility to shift the heat load.

$$FF = \frac{\int_{\text{low price time}} C'_{heating} dt - \int_{\text{high price time}} C''_{heating} dt}{\int_{\text{low price time}} C'_{heating} dt + \int_{\text{high price time}} C''_{heating} dt} \quad (5)$$

where,

$C'_{heating}$ is the energy expense when the price belongs to the low price period.

$C''_{heating}$ is the energy expense when the price belongs to the high price period.

The total heating cost $C_{heating}$:

$$C_{heating} = C_{initial} + C'_{heating} + C''_{heating} \quad (6)$$

$C_{initial}$ is economic cost for the required heat supply at an initial temperature of 28 °C for the filling layer.

3. Result

3.1. Optimal pipe spacing for different DH scenarios

The room temperature with different DH primary supply temperature and pipe spacing is shown in Fig. 8. And Fig. 9 demonstrates RT and the DH charging period for the corresponding cases. Fig. 8 (a) presents the outdoor dry bulb temperature and the solar radiation during the coldest two weeks of the year. Fig. 8 (b) (c) (d) (e) (f) show the room temperature (T_{room}) of the pipe spacings of 100 mm, 200 mm, 300 mm, 400 mm and 500 mm within four DH scenarios (differentiate by the supply temperatures of 35 °C 45 °C, 50 °C and 60 °C) respectively. The occu-

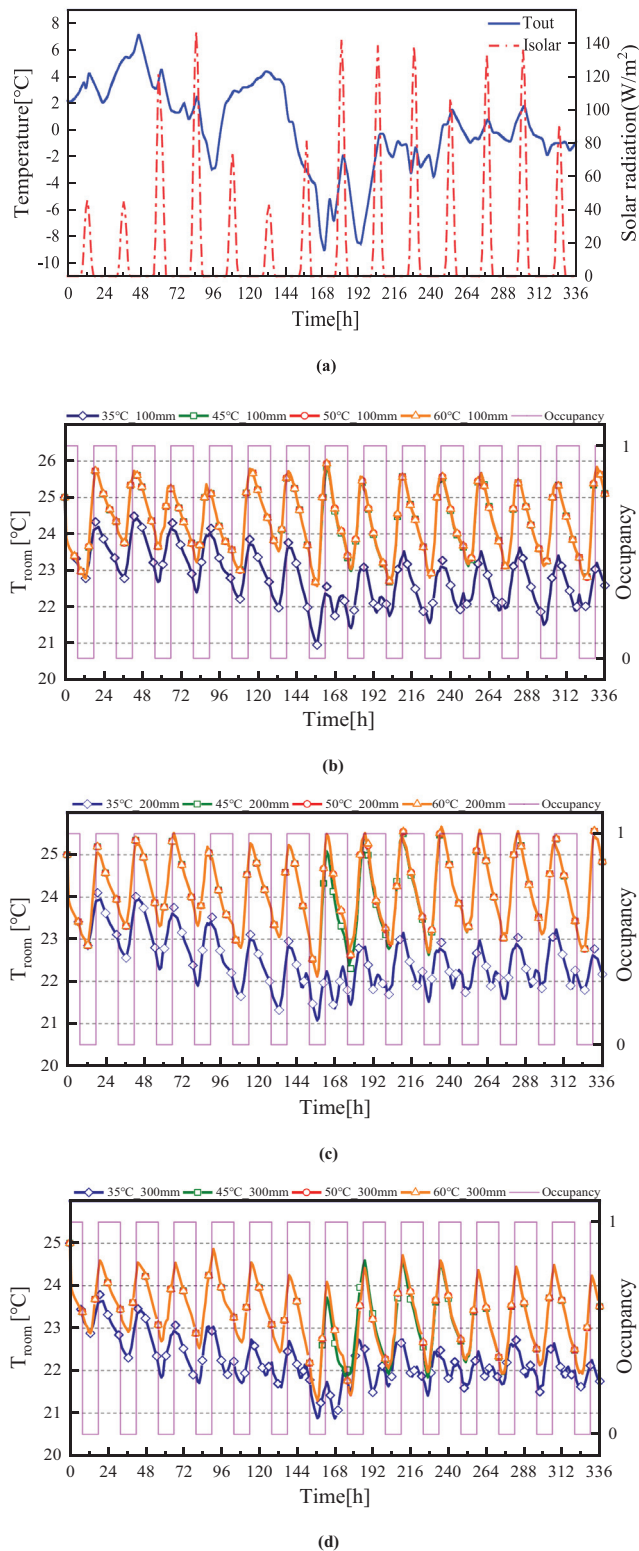


Fig. 8. (a) The weather condition and the room temperature with the pipe spacing of (b) 100 mm, (c) 200 mm, (d) 300 mm, (e) 400 mm and (f) 500 mm at 45 °C, 50 °C and 60 °C of water supply respectively.

pancy represented by the piecewise lines is also shown in the figures.

From the results, the pipe spacing of 100 mm and 200 mm have more serious overheating problem for most of the DH scenarios, except for ULTDH of 35 °C. It occurs mainly at the end of the

charging period for the building mass. Since the setpoint temperature of charging was set to 25 °C to make the utmost use of the low-price energy, the floor filling layer continuously releases heat to the room and causes overheating of the room. Therefore, the relative time of thermal comfort (RT) of 100 mm and 200 mm is less than 1 with the DH supply temperature above 45 °C. The 100 mm pipe spacing scenario show lower RT compared to the scenario of 200 mm pipe spacing due to more severe overheating in Fig. 9 (a).

While for the pipe spacing of 300 mm, 400 mm, 500 mm, the performances are opposite. As the released heat is reduced along with the pipe spacing increasing, the ULTDH with the supply temperature of 35 °C becomes insufficient to provide enough heat to meet the I level of thermal comfort (21 °C – 25 °C). However, for the other three DH scenarios with higher supply temperatures, the overheating problem is resolved by the enlarged pipe spacing. The comfort index of RT of the ULTDH at 35 °C becomes less than 1 for the pipe spacing greater than 300 mm, while the other DH scenarios are all equal to 1. Moreover, as shown in Fig. 9 (b), compared to 300 mm, the pipe spacing of 400 mm, 500 mm requires longer charging time from DH. Therefore, the pipe spacing of 300 mm is the critical level of applying all four DH scenarios with better indoor thermal comfort.

Regarding the effects of the different DH scenarios, the room temperature variations are very similar at the same pipe spacing owing to the reason that the setpoint temperature on the secondary supply temperature is the same, which is 35 °C – 45 °C according to the radiant floor heating design directive. However, it is easier for LTDH and MTDH (with the supply temperature of 50 °C and 60 °C respectively) to achieve the same thermal performance with the identical pipe spacing. Therefore, as shown in Fig. 9 (b), the charging time is two hours longer for the ULTDH at 45 °C compared to the LTDH and MTDH scenarios for the two-week simulation period. To guarantee the thermal comfort of the most critical scenario, the combination of ULTDH at 45 °C and 300 mm pipe spacing are selected for the further analyses in this study.

3.2. Thickness of concrete

3.2.1. Effect of different thickness on heat transfer process

The thickness of the filling layer in the radiant floor has a significant impact on the heat transfer to the room. The variations of the T_{room}, floor upper surface temperature (T_{surf}) and surface heat transfer rate ($\dot{Q}_{shtr}$) of the radiant floor were analyzed for different filling layer thicknesses of 50 mm, 70 mm, 90 mm, 110 mm, 120 mm, 130 mm, 150 mm, 170 mm and 190 mm under the new control strategy. As shown in Fig. 10, the smallest thickness of the filling layer leads to the most dramatic fluctuation of the room air temperature (more than 4 °C). While the room temperature with the 190 mm filling layer fluctuates within the range of 22 °C – 24 °C mostly. It means to increase the concrete core thickness from 50 mm to 190 mm can damp the temperature fluctuation by more than 50% in the case study.

In order to investigate the impact on comfort from different filling layer thickness, the RT values of each case are also calculated. The results are shown in Fig. 11. For the cases of 50 mm, 70 mm and 90 mm, the insufficient building mass results in the significant variation of the indoor air temperature and the overheating problem, which leads to the reduced thermal comfort. Thus, the RT values are all below 1, while the RT value of 50 mm is the lowest. For the cases with the thickness larger than 110 mm, the building mass is sufficient to buffer the overheating or supercooling effect, so the RT values are equaling all to 1.

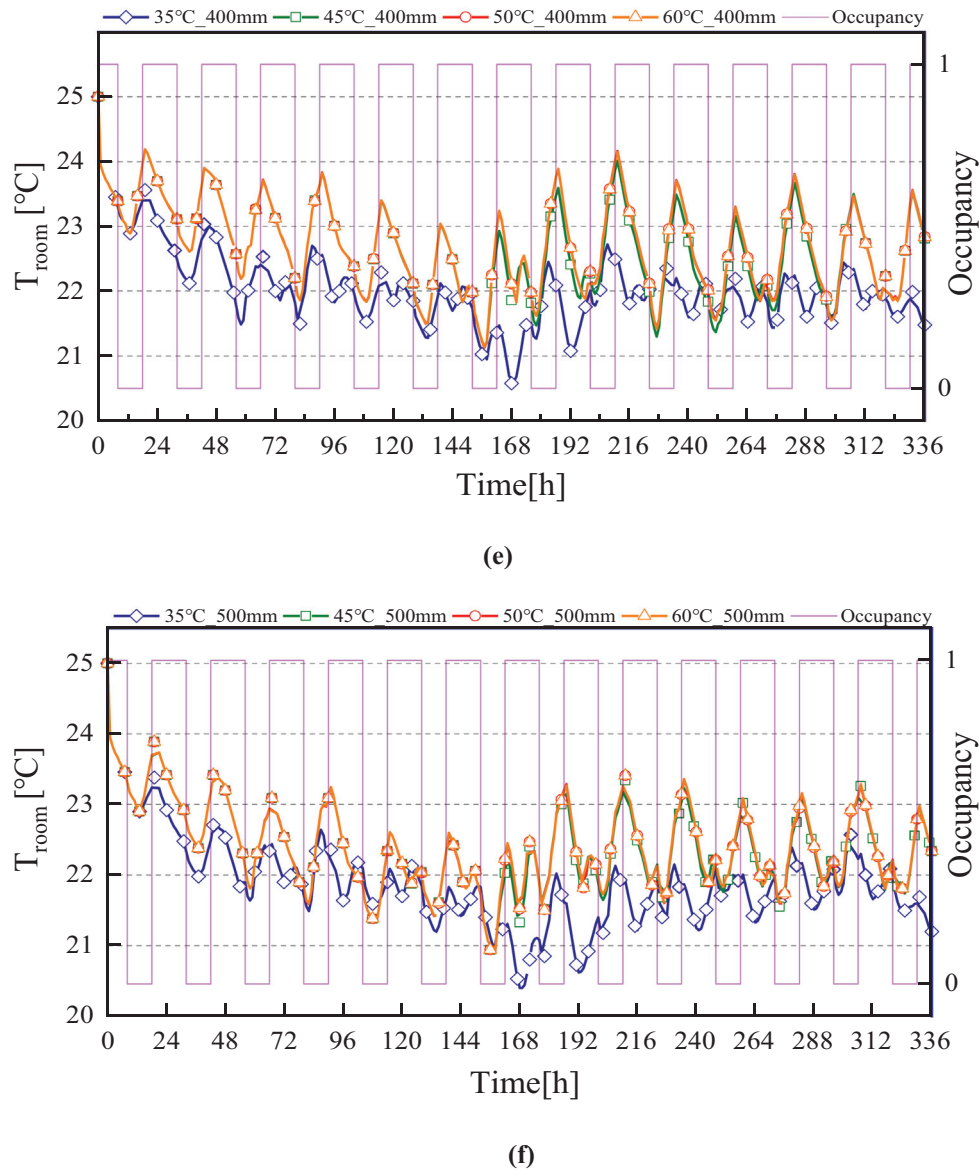


Fig. 8 (continued)

As the important factor that describe the heat transfer between the radiant floor and the indoor environment, T_{surf} and $\dot{Q}_{shtr}$ are investigated. From Fig. 12, the variation of the surface temperature of the radiant floor also greatly depends on the thickness of the concrete core. The surface temperature of the 190 mm concrete core varies between 26 °C – 28 °C approximately, while the largest daily surface temperature variation is around 7 °C for the 50 mm thickness. Moreover, the larger thickness cases can better buffer the thermal input by sudden disturbance from the circumstances, such as the solar radiation.

Fig. 13 represents the $\dot{Q}_{shtr}$ of the radiant floor for different thicknesses of the concrete core. In all the cases, $\dot{Q}_{shtr}$ during the charging period is greater compared to that during the discharging period. But as the concrete thickness increases, the peak value and the variation rate of the $\dot{Q}_{shtr}$ are less significant. After the charging period, the room is heated by the heat released from the building mass until the lower boundary of the setpoint temperature is reached. For the cases with thicker concrete core, the heat stored in the large building mass is sufficient to maintain the I level com-

fort temperature. However, the cases with thinner concrete core thickness requires supplementary heat from the DH, which is represented as the small peak after the discharging period. It can also be noted that the $\dot{Q}_{shtr}$ becomes negative during the mid-day period. This is caused by the fact that the case building has large windows on all four sides, the floor is heated by the strong solar radiation at noon.

3.2.2. Heat flexibility potential and charging/discharging rate

Using the large thermal inertia of the concrete core, the radiant floor can be charged/discharged by regulating the indoor setpoint temperature to realize load shifting. Fig. 14 presents the charging/discharging rate (a) and the heat flexibility potential (b) for different thicknesses of the filling layer. The initial temperature of the filling layer was assumed to 28 °C.

The charging occurs when the occupancy equals 0 and the energy price is low, and the $\dot{Q}_{cr}$ was assumed to be greater than 0. The highest $\dot{Q}_{cr}$ is achieved by the 50 mm case due to the smaller thermal resistance. In contrast, the $\dot{Q}_{dr}$ varied more gently since

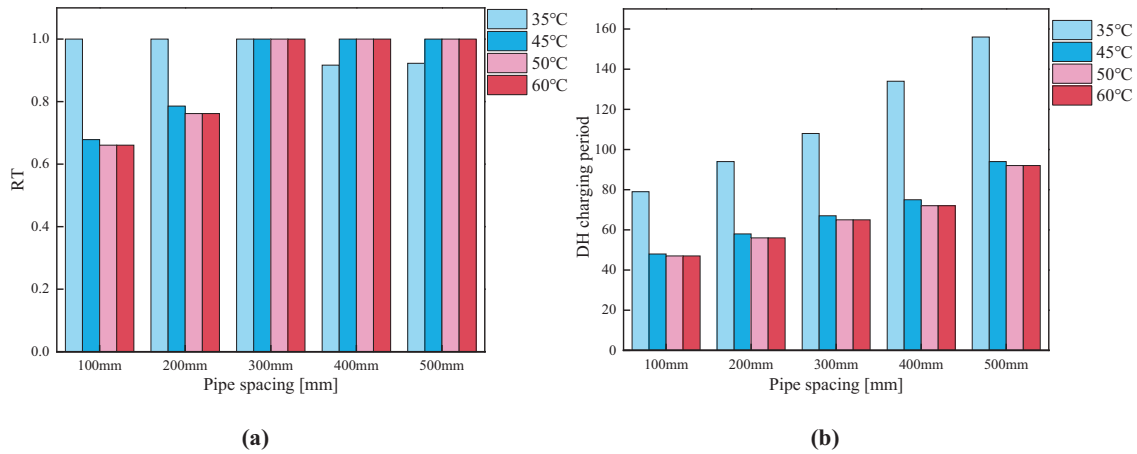


Fig. 9. (a) RT and (b) DH charging period with the pipe spacing of 100 mm, 200 mm, 300 mm, 400 mm and 500 mm at 45 °C, 50 °C and 60 °C of water supply respectively.

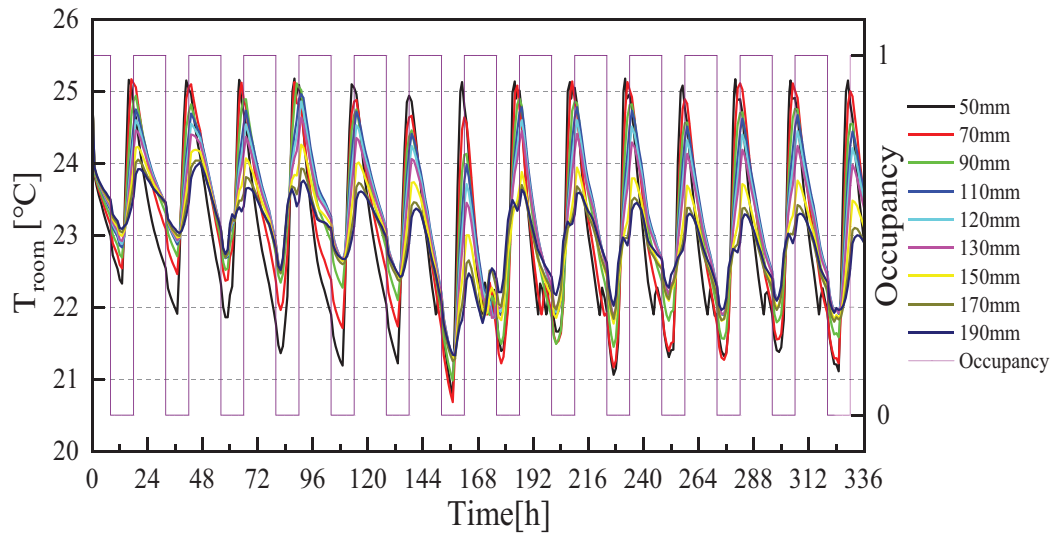


Fig. 10. Room temperature at concrete thickness of 50 mm, 70 mm, 90 mm, 110 mm, 120 mm, 130 mm, 150 mm, 170 mm and 190 mm.

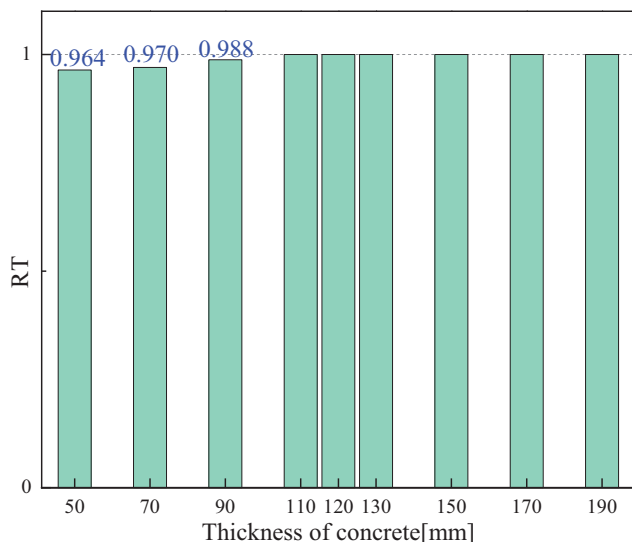


Fig. 11. RT at concrete thickness of 50 mm, 70 mm, 90 mm, 110 mm, 120 mm, 130 mm, 150 mm, 170 mm and 190 mm.

the stored heat is released based on the limited temperature differences between the radiant floor and the indoor air. It also can be seen that as the floor thickness increasing, the $\dot{Q}_{dr}$ decreases more and more slowly. For example, the case of 190 mm shows the more stable $\dot{Q}_{dr}$ compared to the 50 mm case. The larger $\dot{Q}_{dr}$ and the less thermal capacity of the smaller thickness cases shorten the discharging period. Therefore, for the small thickness cases such as 50 mm and 70 mm, the recharging process occurs at late night when the house is supplied by the DH, which can increase the peak of DH supply.

The thermal energy charged to the building mass during the non-occupied period is calculated as the heat flexibility potential, which describes the capability of the building mass for load shifting. From Fig. 14 (b), the highest daily heat flexibility potential is reached by the case of 110 mm. 304.35 kWh of thermal energy can be stored for load shifting in two weeks. The reason is that the thermal storage capacity of the 110 mm case and thicker are sufficient to store the heat from the DH with the prescribed charging rate and the same charging period, while the latter has the larger heat loss due to the large volume and the surface area. For the concrete thinner than 110 mm, the daily heat flexibility potential is lower since the heat capacity is smaller.

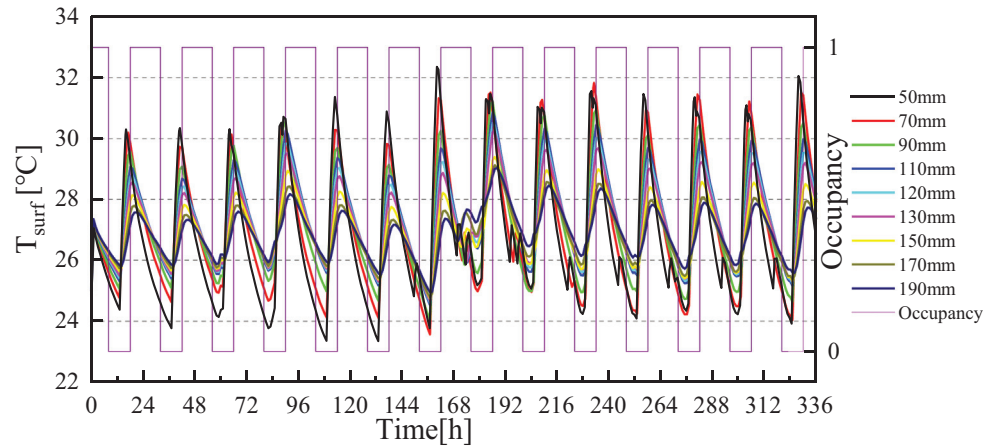


Fig. 12. Floor upper surface temperature at concrete thickness of 50 mm, 70 mm, 90 mm, 110 mm, 120 mm, 130 mm, 150 mm, 170 mm and 190 mm.

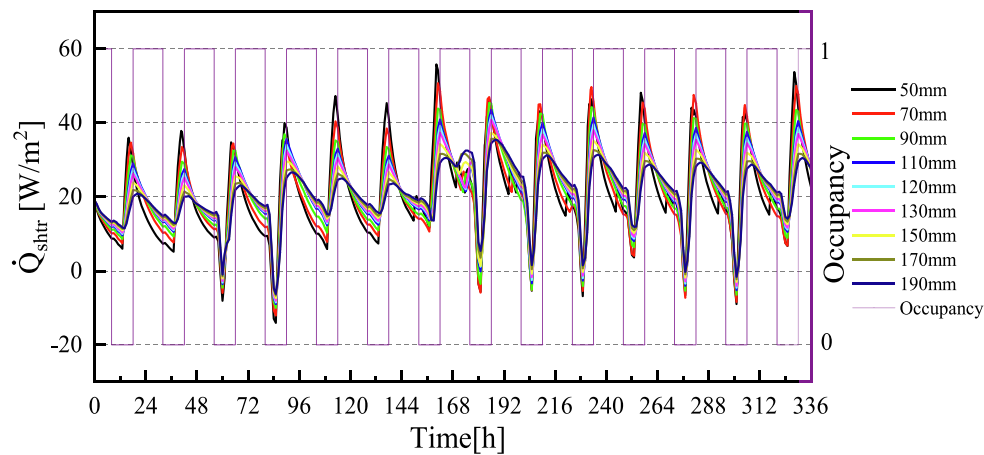


Fig. 13. Surface heat transfer rate of the radiant floor at concrete thickness of 50 mm, 70 mm, 90 mm, 110 mm, 120 mm, 130 mm, 150 mm, 170 mm and 190 mm.

3.2.3. Performance of the scheduled control strategies

From Fig. 15, the peak loads can be completely shifted to the low-price periods with the proposed control strategy. All cases performed well and the flexibility factor (FF) can achieve 1 for all cases, as shown in Fig. 16. The control strategy completely shifts high peak loads to the low demand periods, which eases the pressure of heat and power supply during the peak hours.

Since the heat pump is utilized as the heat source for DH, the energy cost due to the electricity consumption of the heat pump during the two-week period are analyzed. The results are shown in Fig. 17. The C_{heating} is the production of the hourly average electricity price multiplied by the electricity consumption during that period. The total energy cost of all the investigated cases is similar. The maximum cost of electricity is 34.4 DKK when the concrete thickness is 190 mm. While the case of 50 mm has the minimum electricity cost of 31.9 DKK, which is 7 % less.

4. Discussion

The development of LTDH gives more access to the renewable energy sources. Considering the stochastic dynamics of the renewables, it is of great importance to generate and utilize the energy flexibility from both the supply side and the demand side. Under

such context, the energy flexibility provided by the building thermal mass is promising solution. Therefore, we investigated the impact of the configuration parameters of the radiant slab on the thermal performances under different DH scenarios. Different from the conventional continuous operation of the floor heating, scheduled control strategy has to be well designed to drive the charge/discharge process of the building mass, so to activate the building heat flexibility. Accordingly, the stricter requirement to DH is imposed. For example, the supply temperature of 35 °C can be applied to the continuous floor heating, however, it failed to provide enough heat to maintain the thermal comfort with the proposed control strategy for the case building. According to the parametric analysis of the different floor thickness, the 50 mm case achieves better economy with the compromised indoor thermal comfort. The main reason is because that the case building is nearly-zero energy building with very limited heat load. Therefore, it requires a small amount of flexible heat for load shifting. However, in terms of the buildings with lower building code, along with the increase of the heat load, the demand of the building heat flexibility can be larger. It is possible that the radiant floor with smaller thickness does not have sufficient energy flexibility potential, and the recharging has to be performed during the high-price period. Thereby, the results of the economy can be much different.

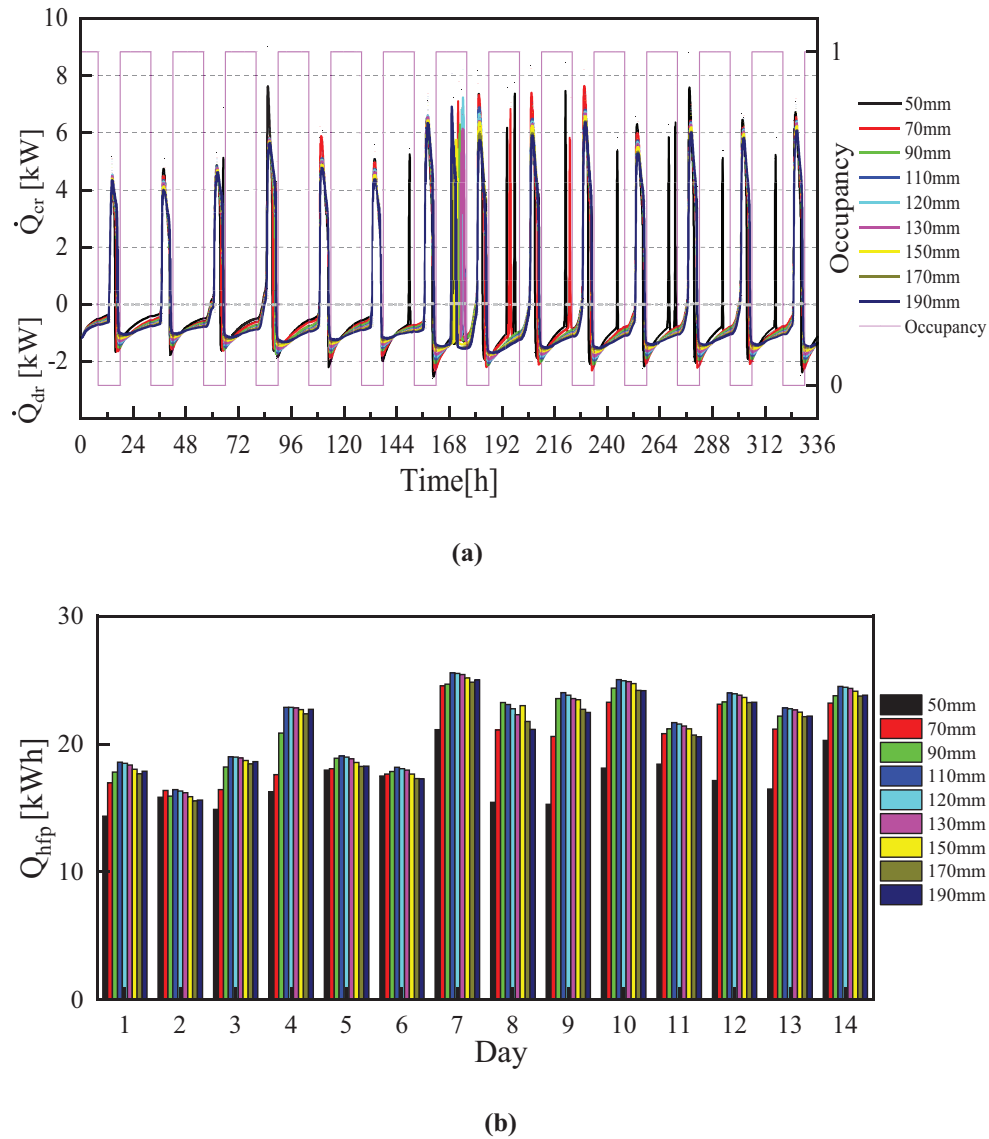


Fig. 14. (a) Charging/discharging rate and (b) daily heat flexibility potential at concrete thickness of 50 mm, 70 mm, 90 mm, 110 mm, 120 mm, 130 mm, 150 mm, 170 mm and 190 mm.

The radiant systems are more suitable for “low-temperature heating and high-temperature cooling” concept. The reason is due to the possibility of the system to be operated with moderate temperature levels, which greatly improves the exergy efficiency by balancing the low-exergy sources with the low-exergy demand [44,45]. Therefore, this study compares the performances of different floor heating configurations under different DH scenarios. The aim is to investigate the critical/optimal design and control parameters that can realize the application of district heating with lower operation temperature. The heat source of this study was represented by a heat pump regarding the actual situation in the case study. Other low-exergy resources, such as the low-intensity renewables, waste heat and etc. are recommended to be coupled with the LTDH/ULTDH, so to reduce the actual CO₂ emission or improve the system efficiency. However, in this study, we put more focus on the energy flexibility potential on the demand side,

and how to quantify and maximize the energy flexibility from the floor heating slab by the optimal control strategy. Therefore, we did not include the concrete exergy analysis in respect of the origin of the energy supply. However, as mentioned by Kilis in [46], in order to implement the exergy efficient radiant heating system, it is of great importance to perform the comprehensive exergy analysis of the system, so to achieve the good match between the source and demand. Moreover, the exergy destruction from the heat exchanger, the pump and other important system components should be included for the exergy evaluation of different DH scenarios in addition to the origin of all the energy supplied to the system. For example, the ULTDH can save the exergy by using the low-exergy source, but it may also result in larger pumping power of electricity compared to the MTDH. Therefore, more concrete analysis on the broader systemic scale is worthy to be further investigated in the future.

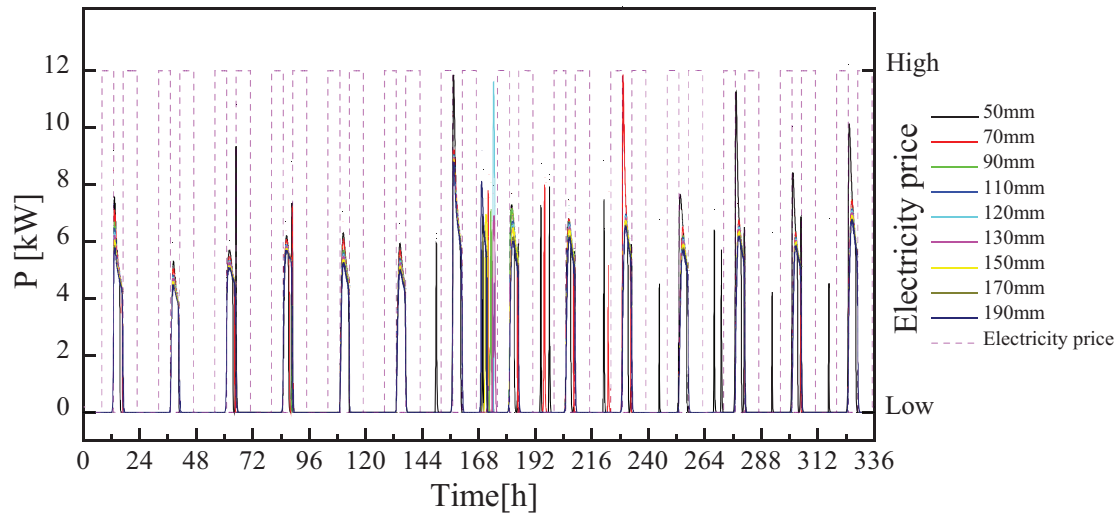


Fig. 15. Power consumption of the heat pump at concrete thickness of 50 mm, 70 mm, 90 mm, 110 mm, 120 mm, 130 mm, 150 mm, 170 mm and 190 mm.

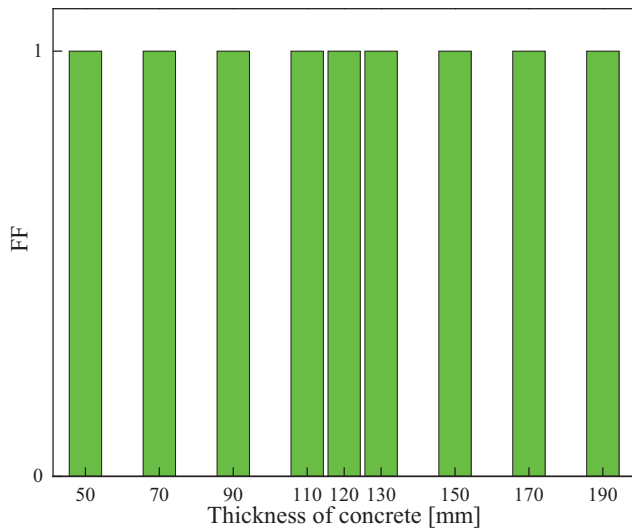


Fig. 16. FF at concrete thickness of 50 mm, 70 mm, 90 mm, 110 mm, 120 mm, 130 mm, 150 mm, 170 mm and 190 mm.

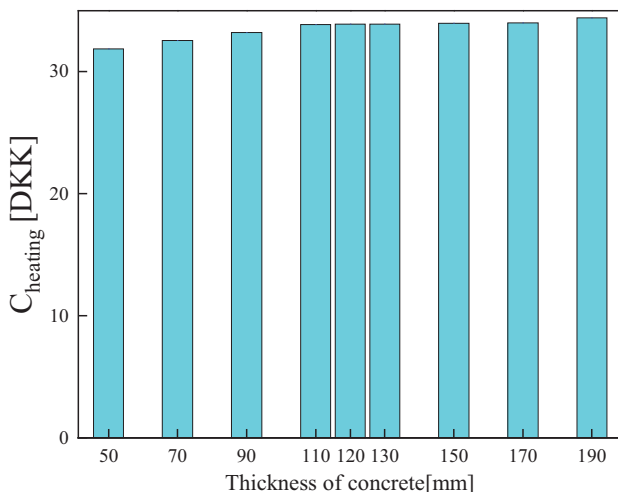


Fig. 17. Electricity Cost at concrete thickness of 50 mm, 70 mm, 90 mm, 110 mm, 120 mm, 130 mm, 150 mm, 170 mm and 190 mm in fourteen days.

5. Conclusion

This article investigates the impacts of the configuration parameters of the radiant floor heating system on the viability of applying different DH scenarios. The aim is to identify the optimal combination of the configuration parameters to realize DH with lower supply temperature. In order to improve the economic and energy performances of the heating system, the scheduled control logic was proposed to regulate load shift by activating the heat flexibility of the building. The conclusion is summarized as follows:

1. At the same DH supply temperature, small pipe spacing can cause overheating problems, while the large pipe spacing leads to longer charging time. The pipe spacing of 300 mm can satisfy all DH scenarios.
2. As the thickness of concrete increasing, the peak load decreases. The surface heat transfer rate and the charging rate become more gentle.
3. All cases can reach the flexibility factor of 1 with the scheduled RBC control strategy. The 110 mm concrete reaches the maximum the amount of total heat flexibility potential in the filling layer, which reaches the 304.35 kWh for the two-week period. While the lower thickness cases achieve better economic performances with compromised thermal comfort.

The results from this study provides knowledge for the design of radiant heat system with LTDH or ULTDH from both the theoretical and practical point of view. The heat flexibility of the building thermal mass is useful to integrate the dynamic heat source and the heat demand. In the future, predictive control will need to be developed to enhance the robustness of the operation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study was carried out with great support and data provided by the EUDP research project Bolig2020, J.nr. 64015-0640.

References

- [1] T. Abergel, J. Dulac, I. Hamilton, M. Jordan, A. Pradeep, Global status report for buildings and construction—towards a zero-emissions, efficient and resilient buildings and construction sector, International Energy Agency, Global Alliance for Buildings and Construction, UN environment programme, (2021).
- [2] P. Taylor, Energy Technology Perspectives, International Energy Agency, <https://www.iea.org/buildings>, 2020.
- [3] B.W. Olesen, Radiant heating and cooling by embedded water-based systems, in: Proceedings of I Congreso Climatización Eficiente, 2011, p. 4.
- [4] L. Zhang, X.-H. Liu, Y. Jiang, Simplified calculation for cooling/heating capacity, surface temperature distribution of radiant floor, Energy Build. 55 (2012) 397–404.
- [5] R. Li, T. Yoshidomi, R. Ooka, B.W. Olesen, Field evaluation of performance of radiant heating/cooling ceiling panel system, Energy Build. 86 (2015) 58–65.
- [6] X. Xu, S. Wang, J. Wang, F. Xiao, Active pipe-embedded structures in buildings for utilizing low-grade energy sources: a review, Energy Build. 42 (10) (2010) 1567–1581.
- [7] O.B. Kazanci, M. Shukuya, A theoretical study of the effects of different heating loads on the exergy performance of water-based and air-based space heating systems in buildings, Energy 238 (2022) 122009.
- [8] M. Ferrara, J. Coleman, F. Meggers, Exploring potentialities of energy-connected buildings: performance assessment of an innovative low-exergy design concept for a building heating supply system, Energy Procedia 122 (2017) 1075–1080.
- [9] S.P. Cognati, A. Kindinis, Thermal mass activation by hollow core slab coupled with night ventilation to reduce summer cooling loads, Build. Environ. 42 (9) (2007) 3285–3297.
- [10] X. Wu, L. Fang, B.W. Olesen, J. Zhao, F. Wang, Comparison of indoor air distribution and thermal environment for different combinations of radiant heating systems with mechanical ventilation systems, Build. Serv. Eng. Res. Technol. 39 (1) (2018) 81–97.
- [11] J. Le Dréau, P. Heiselberg, Sensitivity analysis of the thermal performance of radiant and convective terminals for cooling buildings, Energy Build. 82 (2014) 482–491.
- [12] T. Mikeska, S. Svendsen, Dynamic behavior of radiant cooling system based on capillary tubes in walls made of high performance concrete, Energy Build. 108 (2015) 92–100.
- [13] W. Liao, Y. Luo, J. Peng, D. Wang, C. Yuan, R. Yin, N. Li, Experimental study on energy consumption and thermal environment of radiant ceiling heating system for different types of rooms, Energy 244 (2022) 122555.
- [14] B. Larwa, S. Cesari, M. Bottarelli, Study on thermal performance of a PCM enhanced hydronic radiant floor heating system, Energy 225 (2021) 120245.
- [15] B. Ning, S. Schiavon, F.S. Bauman, A novel classification scheme for design and control of radiant system based on thermal response time, Energy Build. 137 (2017) 38–45.
- [16] M. Krajčák, O. Šikula, Heat storage efficiency and effective thermal output: Indicators of thermal response and output of radiant heating and cooling systems, Energy Build. 229 (2020) 110524.
- [17] Y. Yang, S. Chen, T. Chang, J. Ma, Y. Sun, Uncertainty and global sensitivity analysis on thermal performances of pipe-embedded building envelope in the heating season, Energy Convers. Manage. 244 (2021) 114509.
- [18] D. Wang, Y. Liu, Y. Wang, J. Liu, Numerical and experimental analysis of floor heat storage and release during an intermittent in-slab floor heating process, Appl. Therm. Eng. 62 (2) (2014) 398–406.
- [19] P. Ding, Y. Li, E. Long, Y. Zhang, Q. Liu, Study on heating capacity and heat loss of capillary radiant floor heating systems, Appl. Therm. Eng. 165 (2020) 114618.
- [20] H. Jing, Z. Quan, R. Dong, L. Hao, Y. Liu, Y. Zhao, Performance simulation and optimization of new radiant floor heating based on micro heat pipe array, in: Building Simulation, Springer, 2022, pp. 1–14.
- [21] J. Oravec, O. Šikula, M. Krajčák, M. Anıcı, M. Mohapl, A comparative study on the applicability of six radiant floor, wall, and ceiling heating systems based on thermal performance analysis, J. Build. Eng. 36 (2021) 102133.
- [22] J. Dong, L. Zhang, S. Deng, B. Yang, S. Huang, An experimental study on a novel radiant-convective heating system based on air source heat pump, Energy Build. 158 (2018) 812–821.
- [23] K.M. Luc, R. Li, L. Xu, T.R. Nielsen, J.L. Hensen, Energy flexibility potential of a small district connected to a district heating system, Energy Build. 225 (2020) 110074.
- [24] J. Joe, P. Karava, X. Hou, J. Hu, Model predictive control of a radiant floor cooling system in an office space, (2016).
- [25] B.-C. Ahn, Radiant Floor Heating System, in: Developments in Heat Transfer, Intech, 2011, p. 22.
- [26] L. Zhang, J. Liu, M. Heidarinejad, M.K. Kim, J. Srebric, A two-dimensional numerical analysis for thermal performance of an intermittently operated radiant floor heating system in a transient external climatic condition, Heat Transfer Eng. 41 (9–10) (2020) 825–839.
- [27] T. Li, A. Merabtine, M. Lachi, N. Martaj, R. Bennacer, Experimental study on the thermal comfort in the room equipped with a radiant floor heating system exposed to direct solar radiation, Energy 230 (2021) 120800.
- [28] J. Guo, J. Dong, H. Wang, Y. Jiang, J. Tao, On-site measurement of the thermal performance of a novel ventilated thermal storage heating floor in a NZEB, Build. Environ. 201 (2021) 107993.
- [29] J. Le Dréau, P. Heiselberg, Energy flexibility of residential buildings using short term heat storage in the thermal mass, Energy 111 (2016) 991–1002.
- [30] K. Foteinaki, R. Li, A. Heller, C. Rode, Heating system energy flexibility of low-energy residential buildings, Energy Build. 180 (2018) 95–108.
- [31] H. Lund, S. Werner, R. Wiltshire, S. Svendsen, J.E. Thorsen, F. Hvelplund, B.V. Mathiesen, 4th Generation District Heating (4GDH): Integrating smart thermal grids into future sustainable energy systems, Energy 68 (2014) 1–11.
- [32] B.V. Mathiesen, H. Lund, K. Karlsson, 100% Renewable energy systems, climate mitigation and economic growth, Appl. Energy 88 (2) (2011) 488–501.
- [33] M. Münster, P.E. Morthorst, H.V. Larsen, L. Bregnbæk, J. Werling, H.H. Lindboe, H. Ravn, The role of district heating in the future Danish energy system, Energy 48 (1) (2012) 47–55.
- [34] Executive Order on the Publication of the Danish Building Regulations 2018 (BR18). The Danish Transport and Construction Agency: Danish Building Regulations 2018; 2018.
- [35] Weather Data from EnergyPlus website. <https://energyplus.net/weather>.
- [36] Executive Order on the Publication of the Danish Building Regulations 2015 (BR15). The Danish Transport and Construction Agency: Danish Building Regulations 2015; 2015.
- [37] A. Dalla Rosa, H. Li, S. Svendsen, Method for optimal design of pipes for low-energy district heating, with focus on heat losses, Energy 36 (5) (2011) 2407–2418.
- [38] Dansk Standard, DS/EN ISO 13790. Bygningers energieffektivitet – Beregning af energiforbrug til rumopvarmning og -køling. Energy performance of buildings – Calculation of energy use for space heating and cooling (2.udgave). 2008.
- [39] Chigao heat pump LSCN-20/DF3. <http://www.chigo-geme.com/home-product-productshow-id-295.html>.
- [40] Dansk standard, DS 469:2013. Varme- og køleanlæg i bygninger. Heating and cooling systems in buildings (2.udgave), 2013.
- [41] Dynamic electricity price from the Nord Pool. <https://www.nordpoolgroup.com/>.
- [42] Dansk standard, DS/EN 15251. Input-parametre til indeklimaet ved design og bestemmelse af bygningers energimæssige ydeevne vedrørende indendørs luftkvalitet, termisk miljø, belysning og akustik (1.udgave). 2007.
- [43] S.O. Jensen, A. Marszał-Pomianowska, R. Lollini, W. Pasut, A. Knotzer, P. Engelmann, A. Stafford, G. Reynders, IEA EBC annex 67 energy flexible buildings, Energy Build. 155 (2017) 25–34.
- [44] X. Yang, H. Li, S. Svendsen, Energy, economy and exergy evaluations of the solutions for supplying domestic hot water from low-temperature district heating in Denmark, Energy Convers. Manage. 122 (2016) 142–152.
- [45] X. Yang, S. Svendsen, Ultra-low temperature district heating system with central heat pump and local boosters for low-heat-density area: Analyses on a real case in Denmark, Energy 159 (2018) 243–251.
- [46] B. Kilis, Exergy metrication of radiant panel heating and cooling with heat pumps, Energy Convers. Manage. 63 (2012) 218–224.